

Abstract

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Chapter 1

Strategic Vision and Project Scope

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1.1 General Introduction

Digital transformation has changed how educational institutions manage teaching, communication, assessment, and academic follow-up. Despite this shift, many institutions still rely on fragmented combinations of messaging applications, file-sharing tools, video-conferencing platforms, and manually coordinated assessment processes. Such fragmentation reduces continuity in the learning experience, increases administrative overhead, and weakens the institutional ability to manage educational workflows within one controlled environment.

Eduverse is an integrated educational platform developed to address this problem through a unified, multi-organization, role-based environment. Its main implemented client is a web application, supported by an MVP mobile companion application that extends access to selected academic workflows for students and teachers. The platform is centered on the class as the primary digital workspace in which materials, assignments, examinations, results, notifications, live sessions, academic communication, and intelligent support services can be managed in a coordinated manner. Within this workspace, classes are organized by term and stage, can move to Past Terms when concluded, and preserve academic history under organization-level visibility and feature controls.

At a high level, Eduverse combines a modern web platform with supporting cloud and real-time services in order to provide a realistic and extensible educational system. The web client is built using Next.js, React, TypeScript, Tailwind CSS, and Supabase, while the mobile companion is built using Expo, React Native, and TypeScript. These technologies are mentioned here only as an introductory overview; detailed technical analysis and implementation discussion are reserved for later chapters.

1.2 Problem Statement

Educational institutions often manage closely related academic tasks through separate systems that are not designed to operate as one coherent platform. Materials may be distributed through one service, communication may take place in another, live teaching may depend on a third platform, and assessment records may be maintained elsewhere. As a result, students are forced to move between disconnected interfaces, while teachers and

administrators struggle to maintain consistent oversight of participation, submissions, communication, and evaluation.

This problem becomes more serious in institutions that must manage multiple organizations, classes, and user roles under different access conditions. A practical educational platform must support not only learning activities, but also institutional control, membership management, and role-based access across different organizational contexts. Without these capabilities, digital education systems tend either to become rigid and limited or to become too loosely structured to support reliable academic administration.

A further challenge concerns platform coverage across devices. Rich authoring, administration, and class management workflows are best handled through a full web environment, whereas frequent day-to-day actions such as checking classes, following notifications, viewing assignments, opening materials, joining live sessions, and reading messages must also be accessible on mobile devices. This creates a need for an educational platform centered on a comprehensive web environment, supported by a practical mobile companion rather than a simplistic one-client-for-all approach.

1.3 Project Motivations

The development of Eduverse is motivated by the need to provide a unified academic environment in which essential educational workflows can be carried out without fragmentation. A coherent class-centered system can improve the student experience, reduce operational burden for teachers and administrators, and strengthen institutional visibility over communication, assessment, and academic progress.

The project is also motivated by the growing demand for flexible educational infrastructure. Modern institutions often require platforms that can support multiple organizations, distinct user roles, synchronous and asynchronous learning activities, and selective feature enablement within one extensible environment. Eduverse responds to this need by combining institutional control with practical support for daily teaching and learning workflows.

From an academic perspective, Eduverse is also motivated by its value as a software engineering project. It provides a realistic setting in which requirements analysis, interface planning, system integration, data

management, real-time communication, AI-assisted services, and technical documentation can be studied and implemented within one coherent graduation project context.

1.4 Strategic Objectives

At the strategic level, Eduverse aims to establish an integrated educational platform whose primary implemented client is a web application, complemented by an MVP mobile companion application, in order to centralize essential academic workflows for organizations, teachers, and students within one role-aware and extensible system.

The main strategic objectives of the project are as follows:

- To centralize core educational workflows such as class management, materials, assignments, examinations, results, notifications, communication, and live sessions within one coordinated platform.
- To support multi-organization operation and role-based access for organization administrators, teachers, and students, including memberships spanning multiple organizations or roles with active organization and role switching according to the current workspace context.
- To provide a primary web-based environment capable of supporting both institutional administration and day-to-day academic activity.
- To extend selected high-frequency academic workflows to a mobile MVP companion application without claiming full feature parity with the web platform.
- To incorporate distinctive educational capabilities such as an AI Agent and an integrated coding workspace as part of the platform's broader academic value.
- To organize data and services in a maintainable and scalable manner suitable for educational institutions with different operational needs.
- To produce a graduation project that is both academically meaningful and practically relevant as a modern educational software platform.

1.5 Educational and Professional Goals for the Team

Eduverse also serves clear educational and professional goals for the project team. From an educational standpoint, it provides an opportunity to apply software engineering concepts to a realistic problem domain rather than to an isolated classroom exercise. The team must analyze institutional needs, define system boundaries, organize requirements, structure user roles, and translate these decisions into a functional digital platform.

From a professional standpoint, the project strengthens practical experience in requirements analysis, frontend development, backend integration, database usage, real-time service integration, collaborative development workflows, and technical documentation. It therefore contributes not only to the production of software, but also to the development of disciplined engineering practice, teamwork, and academic reporting skills.

1.6 Project Scope and Boundaries

The scope of Eduverse in this dissertation is defined around the implemented web platform and the existing mobile MVP companion application. Chapter 1 is limited to introducing the problem context, strategic direction, main user groups, system boundaries, and general dissertation structure. Detailed requirements, architecture, database design, implementation decisions, and testing evidence are reserved for later chapters.

1.6.1 System Boundaries and User Roles

Eduverse operates within a multi-organization educational context and therefore distinguishes clearly between administrative, instructional, and learner responsibilities. At the institutional level, the platform supports organization administrators who manage organizations, user access, public and invitation-based membership onboarding, public join links, approval-based join requests, feature configuration, class oversight, teacher permission controls, and operational control. A user may belong to multiple organizations, and the same organization member may hold more than one active role, with the interface and permissions resolved according to

the currently selected organization and role. At the instructional level, teachers manage class activities such as materials, assignments, live sessions, examinations, and academic follow-up within the permissions defined by organization settings. At the learner level, students access the classes available to them in order to study materials, follow announcements, submit work, join sessions, communicate, and review academic outcomes.

These role distinctions define important system boundaries. Eduverse is not a generic social platform in which all users have the same functional position. Instead, access is governed by organizational membership, class context, and academic responsibility. This role-aware structure supports accountability, controlled visibility, and more reliable institutional management.

1.6.2 Functional Scope

Within its current implemented scope, Eduverse includes authentication, organization access and multi-organization membership, class management, materials, assignments, exams, results and academic history, notifications, live sessions, an AI Agent, and an integrated coding workspace. Together, these functions define the platform as more than a basic learning content portal. Eduverse is intended to support both academic management and active instructional workflows within one system. Within this scope, classes can be visible across the organization, require term and stage definition, and move into Past Terms without losing preserved academic history. Teacher management abilities, class feature controls, student linkage to previous terms, and Results visibility are also governed through organization and class settings.

At the same time, Chapter 1 remains introductory in nature. It does not attempt to document detailed database structures, API behavior, or low-level internal logic. Those aspects are part of the later analytical and technical chapters and lie outside the present scope.

1.6.3 Web and Mobile Client Scope

The main implemented Eduverse client is the web application. It provides the widest operational coverage of the platform, including organization-aware navigation, role-based access, membership handling, class management, materials, assignments, exams, results, notifications, live sessions, AI Agent

workflows, and the integrated coding workspace. For this reason, the web application should be regarded as the core working environment of the system.

The mobile application has a more focused scope. It is an MVP companion application intended to extend access to selected academic workflows such as authentication, dashboard viewing, classes, assignments, notifications, chat, materials, and live sessions. Although it is connected to the same broader platform environment, it is not yet feature-complete relative to the web application and should therefore be described as a companion application rather than a replacement for the primary platform.

1.6.4 Competitive Features and Technological Innovation

Eduverse is distinguished by the combination of features it brings together in one educational platform. A major distinguishing point is its multi-organization, role-based structure, which allows the same system to support organizational control, instructional management, and student participation in a coordinated way. The platform also integrates live sessions, an AI Agent, and a coding workspace within the same broader academic environment rather than treating them as isolated external utilities. It also supports controlled public and invitation-based class access, term-and-stage class organization, and the transition of completed classes into Past Terms while preserving historical records.

Another important feature is the platform's attention to examinations and assessment integrity. Eduverse does not treat evaluation as a peripheral task, but as an integral part of the educational workflow that must be managed within a controlled digital environment, including controlled student-facing Results visibility where needed. In addition, the system deliberately combines a comprehensive web platform with an MVP mobile companion application, thereby balancing rich institutional workflows with accessible everyday mobile use.

From a technological perspective, Eduverse also reflects a deliberate service separation strategy. Structured and frequently queried academic data are stored through Supabase PostgreSQL, large educational files are handled through AWS S3, and real-time live-session interaction is supported through LiveKit. This separated storage and service model strengthens the

platform's practicality for an educational environment that must manage both relational records and larger real-time or file-based workloads.

1.7 Report Organization

This dissertation is organized as follows. Chapter 2 presents the literature review and theoretical framework. Chapter 3 describes the adopted methodology. Chapter 4 covers system analysis and requirements. Chapter 5 presents system design. Chapter 6 discusses software and engineering implementation. Chapter 7 presents testing and evaluation. Chapter 8 addresses future work and planned extensions, followed by the conclusion and appendices.

Chapter 2

Literature Review and Theoretical Framework

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2.1 Introduction

This chapter presents the theoretical background for Eduverse. It reviews the evolution of digital learning systems, the limitations of fragmented educational tools, selected concepts from artificial intelligence in education, and the main foundations that inform the system's design. The discussion draws selectively from the literature on Learning Management Systems, Artificial Intelligence in Education, Role-Based Access Control, and educational platform architecture.

2.2 Evolution of Digital Education Platforms

2.2.1 From Traditional Classroom Management to Digital Learning Systems

Educational management has historically depended on face-to-face coordination, manual attendance records, printed materials, and locally maintained assessment processes. As educational institutions expanded in scale and digital communication became more common, Learning Management Systems emerged to centralize course materials, announcements, assignment handling, and academic tracking within a unified digital environment [1, 2].

The move toward digital learning systems did not simply replace paper-based administration with electronic storage. It introduced new expectations about continuous access, asynchronous participation, online communication, and traceable academic workflows. Modern educational platforms are therefore judged not only by their capacity to store materials, but also by their ability to support coordination, interaction, assessment, and institutional oversight across different users and contexts [11].

2.2.2 Limitations of Fragmented Learning Management Tools

Although digital tools have become common in education, many institutions still depend on separate services for class management, communication, file

sharing, video sessions, assessment, and student support. This fragmented model creates operational discontinuity because activities that belong to the same academic process are distributed across unrelated systems. As a result, students must move repeatedly between platforms, while teachers and administrators face a reduced ability to maintain clear academic oversight [11, 2].

When communication, submissions, live sessions, and evaluation records are distributed across multiple tools, consistency, accountability, and access governance become harder to maintain. The limitation is therefore organizational as well as technical, because it weakens the integrity of the educational environment itself.

2.3 Research Gap and Proposed Integrated Learning Ecosystem

A clear gap exists between the broad range of activities required in modern education and the way those activities are often supported by disconnected systems. Institutions may rely on one tool for class coordination, another for messaging, another for live teaching, another for assignments, and still others for assessment, AI support, or programming practice. While each tool may be effective in isolation, the overall experience remains fragmented and operationally inefficient.

Eduverse addresses this gap by proposing an integrated learning ecosystem rather than a collection of separate utilities. It brings class management, communication, materials, assessment, live sessions, AI-assisted support, and programming practice into one organizationally governed environment. The main implemented client is the web application, while the mobile client remains an MVP companion for selected daily-use workflows. This balance between integration and controlled scope is central to the academic identity of the project. It also reflects the need for one platform to coordinate controlled public access, role-aware governance, and preserved academic history across active and archived class terms.

2.4 Artificial Intelligence in Education

Artificial intelligence in education is commonly discussed as a means of augmenting teaching and learning rather than replacing teachers or institutional judgment [8, 6]. In educational settings, AI systems are valuable when they improve access to explanations, support learning continuity, reduce routine workload, and provide structured assistance without undermining academic responsibility. In Eduverse, this perspective frames the AI Agent as a bounded academic support service embedded in the platform rather than as an authority over teaching or evaluation.

2.4.1 Context-Aware Learning Assistance

One of the most important ideas in AI-supported education is that assistance is more useful when it is tied to the learner’s actual academic context. Generic responses are often less effective than support that takes into account course materials, class discussion, ongoing tasks, and instructional expectations. Context-aware assistance can therefore improve relevance, reduce confusion, and support more focused academic interaction [6].

In Eduverse, this means the AI Agent operates within the active class, user role, and immediate learning task so that support remains relevant to the current course activity. Its role is to assist understanding and clarification rather than to substitute for teaching.

2.4.2 Material Summarization and Learning Support

Material summarization represents a practical form of AI assistance in education. Long documents, learning materials, and instructional resources can be difficult for students to process efficiently, especially when they are working under time constraints or trying to review several academic topics in parallel. AI-assisted summarization can support comprehension by highlighting main points, reducing information overload, and helping learners return more quickly to the central ideas of a topic [8].

This kind of support is most appropriate when it complements reading rather than replacing it. In Eduverse, it is most relevant to class materials and related course resources, where AI-generated summaries can complement class-based AI interaction while keeping the original material central.

2.4.3 Assessment and Feedback Support

AI can also support assessment-related workflows by assisting with draft guidance, formative feedback, and structured academic support around tasks and examinations. In theory, such support can help learners prepare, reflect on their work, and identify areas that require improvement [6]. However, educational literature also emphasizes that AI use in assessment must be governed carefully so that it supports learning without weakening academic integrity or teacher responsibility [8, 6].

For Eduverse, this means the AI Agent may assist students and teachers with preparatory guidance, clarification, and formative support around assignments and examinations, while final instructional judgment and assessment responsibility remain with the teacher. During active assessed work, such support must remain bounded and should not provide direct final answers in place of genuine student effort. This governance also extends to controlled student-facing Results visibility as part of assessment integrity.

2.5 Benchmarking: Fragmented Tools vs. Integrated Eduverse Approach

Table 2.1 summarizes the difference between a fragmented educational-tool model and the integrated approach proposed by Eduverse.

2.6 Theoretical and Technical Foundations of Eduverse

2.6.1 Web-Based Educational Platforms

Web-based educational platforms remain the strongest medium for full academic management because they support richer interfaces, broader administrative workflows, and deeper integration across organizational functions. They are especially suitable for tasks such as class configuration, materials management, assessment setup, and detailed instructional coordination.

From a theoretical standpoint, web environments are architecturally well suited to full academic management because they can centralize complex

Table 2.1. Fragmented Educational Tools vs. Integrated Eduverse Approach

Educational Need	Fragmented Tool Approach	Integrated Eduverse Approach
Class management	Class records, memberships, and academic coordination are handled through separate administrative tools or partially manual processes.	Class organization is managed within one platform through organization-aware and role-based structures.
Communication	Messaging and announcements are distributed across external chat or notification tools.	Communication remains connected to the academic environment and class context.
Live sessions	Synchronous teaching depends on external meeting services disconnected from course workflows.	Live sessions are treated as part of the broader educational platform and linked to class participation.
Materials and assignments	Learning resources and tasks are often stored, shared, and submitted through different services.	Materials and assignments are coordinated within the same academic workspace.
AI support	AI tools are usually external and not directly tied to the learner's course context.	AI support is positioned within the class environment as contextual academic assistance.
Programming practice	Coding activities, when needed, often require separate platforms or disconnected lab tools.	Programming practice can be supported through an integrated coding workspace within the platform ecosystem.
Assessment	Examinations and grading may rely on detached tools with limited continuity or oversight.	Assessment workflows are integrated into the same role-aware educational system.

interfaces, persistent records, administrative controls, and cross-workflow coordination within one accessible system.

2.6.2 Mobile Companion Learning Applications

Mobile educational applications play a different but still important role in digital learning environments. Their strength lies in quick access, timely participation, and support for routine high-frequency actions such as checking notifications, reviewing assignments, opening materials, and joining live sessions. As a result, they are especially useful as companion systems for daily academic engagement.

In Eduverse, the mobile client should therefore be understood as an MVP companion rather than as a complete replacement for the main platform. It extends accessibility to core user-facing workflows while more comprehensive management and authoring remain centered in the web application.

2.6.3 Real-Time Virtual Classrooms

Synchronous learning environments are an important part of modern educational theory because they strengthen immediacy, interaction, and instructional presence. When live classrooms are integrated into the broader academic platform rather than isolated in external services, continuity between teaching, communication, and follow-up becomes easier to maintain [5].

This foundation is relevant to Eduverse because live sessions are not treated as a separate product layer. Instead, they are part of the integrated learning environment and serve the broader goal of linking synchronous participation with the rest of the class workflow, including session chat, presentation annotation, and controlled whiteboard participation.

2.6.4 Integrated Coding Environments for Programming Education

Programming education often requires an environment in which learners can write code, inspect results, practice problem solving, and remain connected to the instructional context of the class. When coding practice is separated entirely from the educational platform, students may lose continuity between learning materials, teacher guidance, assignments, and practical execution.

An integrated coding environment addresses this problem by placing programming practice within the same educational ecosystem as the rest of the academic workflow. For Eduverse, this direction is interpreted pragmatically as a browser-safe coding workspace with guided runners and extension-launch support rather than as a full backend IDE or unrestricted backend execution environment.

2.6.5 Security, Roles, and Data Management

Trustworthy educational platforms depend on controlled access, clear role distinctions, and dependable data management. Role-Based Access Control is especially relevant in this context because it supports structured separation between administrative, teaching, and learner permissions [10, 4]. In educational systems, such distinctions are necessary not only for security, but also for accountability and appropriate institutional governance. In Eduverse, this perspective also supports multi-role memberships in which access is resolved according to the active role within the current organization. In broader institutional terms, this kind of governance also aligns with active-role selection, controlled public access requests, and permission-scoped teacher management.

Data management is equally foundational. Educational systems must maintain reliable records, protect assessment-related information, and support different types of content ranging from structured academic data to larger learning materials and live-session interaction. In Eduverse, this also supports separation between relational academic records and larger instructional files so that storage handling, permissions, and delivery can be controlled more appropriately. It also supports the preservation of historical class records across archived terms so that academic continuity does not depend only on the active class state. From a theoretical perspective, this requires a design orientation that treats security, data organization, and permissions as core educational infrastructure rather than as secondary technical concerns. This view is consistent with Eduverse, where access control, data separation, and institutional structure are part of the platform's trust model.

Taken together, these theoretical threads show that effective educational platforms must reduce fragmentation while coordinating complementary forms of support across web, mobile, synchronous, and practice-oriented learning contexts. They also indicate that AI assistance, real-time

instruction, integrated coding environments, and secure data management are most valuable when they operate within one coherent academic system rather than as detached services. For Eduverse, this combined perspective provides the conceptual basis for a unified platform model with differentiated client roles, bounded AI support, live-session continuity, and protected institutional workflows. The following chapters build on this foundation to explain the analysis, design, and implementation choices of the project.

Chapter 3

Methodology

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3.1 Introduction

This chapter explains the methodology adopted for planning, developing, validating, and refining Eduverse as a graduation project. It focuses on the practical development approach and workflow structure rather than on detailed implementation artifacts, which are discussed later in the thesis.

Eduverse required repeated refinement because it combines the web application as the primary client with an MVP mobile companion, backend services, database migrations, role-based workflows, live sessions, AI-assisted features, and an IDE-based coding workspace.

3.2 System Development Methodology

Eduverse is best described as having been developed through a hybrid iterative workflow with predominantly feature-based vertical slices. This description is more accurate than a strictly phase-based model because the repository evidence shows repeated feature increments rather than a simple sequence such as frontend first, backend later, and testing only at the end. After an initial prototype and project setup stage, the system progressed through iterative increments in which the selected module was implemented, reviewed, and refined in relation to the wider platform.

The methodology remained incremental, but it was not organized as isolated technical layers. When a feature required server-side behavior, permission updates, or data-model changes, those concerns were usually developed close to the same increment rather than postponed to a later project phase. Within this process, the web application remained the primary execution environment, while the mobile application was kept as an MVP companion for selected high-frequency academic activities.

3.2.1 Feature-Based Iterative Workflow

Development generally proceeded through working slices organized around real academic or administrative workflows. Examples include organization management, class workflows, materials, assignments, examinations, live sessions, IDE support, and AI-assisted features. In many increments, the selected slice combined interface work with the corresponding API or server logic, shared domain services, and Supabase migrations when required.

This pattern allowed the project to evolve through usable increments rather than through disconnected partial implementations. At the same time, the repository evidence also shows refactoring, infrastructure work, and later refinements, so the methodology is most accurately described as predominantly feature-based rather than exclusively vertical-slice in every commit.

3.2.2 Justification for Methodology Selection

This methodology suited Eduverse because the platform combines a multi-organization structure, role-based access, and diverse academic workflows with recurring integration between interface behavior, backend logic, and data constraints. A rigid linear model would have made it harder to adjust permissions, workflow boundaries, and feature integration as requirements became clearer through implementation.

It also supported continuous validation. As features evolved, the project needed repeated confirmation that interface behavior, backend logic, permissions, schema changes, and shared services remained coherent. This was especially important because some increments affected multiple workflows at the same time.

3.3 Development Workflow Structure

In practical terms, the workflow followed a recurring structure in which the team selected a feature or workflow, implemented the required slice, validated it, and integrated it into the main project history.

Repository evidence also shows that build, type-check, and format-check validation was configured through GitHub Actions for pull requests and updates to the main branch. In practice, validation also included targeted `bun test` usage for selected higher-risk logic where such tests were available, together with Biome-backed formatting support. The repository does not support a claim of universal automated test coverage. Detailed CI/CD behavior, deployment setup, and broader engineering artifacts are therefore left for Chapter 6. Accordingly, this chapter describes the methodology as a controlled workflow for iterative feature integration rather than as a set of isolated development phases.

Table 3.1. Eduverse Development Workflow Structure

Workflow step	Main focus
Feature selection and scope clarification	Choose the next feature or module increment and define the user-facing workflow to be completed within the current scope.
Slice implementation	Build the relevant interface behavior and supporting module code for the selected workflow.
Backend and domain integration	Add API routes, server logic, and shared services when the feature depends on server-owned behavior or shared rules.
Data-model refinement	Add Supabase migrations or policy changes when the feature changes permissions, workflow state, or stored academic records.
Validation	Use build, type-check, and format-check routines together with targeted tests for selected high-risk logic where such tests were available.
Integration	Integrate verified changes into the main project history and continue refinement through later increments when needed.

Chapter 4

System Analysis and Requirements

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4.1 Introduction

This chapter presents the system analysis of Eduverse by defining the operational problem space, summarizing the adopted analysis method, organizing the functional and non-functional requirements, and documenting the use-case and activity models that explain system behavior.

4.2 Problem Domain and Operational Gaps

Eduverse addresses the operational gap that emerges when educational institutions rely on separate tools for class administration, communication, learning materials, assignments, examinations, live sessions, and results. Such fragmentation weakens continuity across the learning process, increases context switching for teachers and students, and makes it difficult to govern academic workflows within one role-aware institutional environment.

The problem is intensified by multi-organization operation and differentiated responsibilities, which require controlled membership, permissions, and visibility boundaries. The same environment must also accommodate AI-assisted learning support and programming support without relying on detached external services.

4.3 Analysis Methodology Adopted

The analysis adopted for Eduverse was based on the implemented and validated project scope rather than on speculative features. It proceeded through the following activities:

- reviewing the implemented web platform and the MVP mobile companion in order to identify the actual service boundaries;
- identifying the needs of the organization owner, teacher, and student as the principal stakeholders;
- organizing recurring educational workflows into functional modules that reflect the operational structure; and
- modeling system behavior through use-case and activity diagrams to relate scope, actors, and workflow progression.

4.4 As-Is Analysis

Before Eduverse, institutions commonly combine messaging applications, file-sharing services, live-meeting tools, assessment utilities, and manual follow-up practices. This arrangement separates class communication from materials, submissions, results, coding practice, and contextual support, which produces inconsistent records and slows coordination across roles.

4.5 To-Be Analysis

Eduverse proposes a unified, role-aware environment centered on the class context and coordinated through organization membership and controlled access. The web application serves as the full management environment, while the mobile application remains an MVP companion for selected daily actions; automated notifications, AI support, and IDE support are integrated without expanding the mobile scope beyond that role.

4.6 Requirements Engineering

The requirements of Eduverse are derived from the problem-domain analysis, the validated project scope, and the needs of the principal stakeholders. They are stated at the module level so that the system scope is clear without repeating every action already represented in the use-case diagrams.

4.6.1 Functional Requirements

- **A. Authentication and Access:** Eduverse shall support public entry, account registration, login and logout, verification-email dispatch, and organization or role selection when multiple eligible contexts exist.
- **B. Organization and Role Management:** the platform shall allow organization owners to create organizations, apply feature presets, manage members and invitations, and assign or remove roles.
- **C. Class Management:** the platform shall support class creation, update, teacher assignment, membership control, and feature configuration.

- **D. Course Content and Materials:** the platform shall support material upload with metadata, viewing, search, filtering, download, and controlled deletion.
- **E. Class Communication and Notifications:** the platform shall provide class messaging and event-driven notifications with viewing, read-state management, and follow-up actions.
- **F. Assignment Management:** the platform shall support assignment authoring, publication, student submission through the allowed response modes, instructor grading, and feedback visibility.
- **G. Exam Management and Integrity:** the platform shall support exam authoring, optional passcodes, student attempts, integrity monitoring, review decisions, grading, and optional result release.
- **H. Results and Dashboards:** the platform shall provide role-specific dashboards and controlled access to class, assignment, and examination results, including detailed views where permitted.
- **I. Live Sessions and Whiteboard:** the platform shall allow teachers to start and end sessions and eligible participants to join, communicate, and use whiteboard features.
- **J. AI Learning Support:** the platform shall provide class-context AI assistance for explanation, summarization, drafting support, and bounded assessment-related support without replacing teacher judgment.
- **K. Integrated Coding Lab / IDE Extension:** the platform shall support environment selection, file operations, code editing, execution, terminal use, preview, and problem or console viewing for programming-oriented learning.
- **L. Profile and Help:** the platform shall allow users to manage profile-level preferences, switch active roles where permitted, and consult searchable help content.

4.6.2 Non-Functional Requirements

- **Security and role-based access control:** access to organizations, classes, files, assessments, and live-session resources shall remain protected through authenticated, permission-controlled use.
- **Privacy:** academic records, messages, submissions, and results shall be handled with restricted visibility.
- **Reliability:** the platform shall preserve correct state for memberships, class records, materials, submissions, and examination data.
- **Usability:** core workflows shall remain understandable to organization owners, teachers, and students without unnecessary context switching.
- **Real-time responsiveness:** live sessions, chat, notifications, and related synchronous interactions shall provide timely updates.
- **Maintainability:** the system shall remain modular enough to support continued evolution of features such as AI support, live sessions, and IDE functionality.
- **Scalability:** the platform shall remain suitable for multiple organizations, multiple classes, and growing volumes of academic records and uploaded files.
- **Cross-client compatibility:** the mobile companion shall remain consistent with the permission and data model of the web platform while preserving its narrower scope.
- **Performance:** dashboards, files, results views, and real-time features shall perform at a level appropriate for regular educational operation.

4.7 Use Case Modeling

Use-case modeling expresses Eduverse through the interactions between the principal actors and the services available to them. The diagrams are the main analytical artifacts in this section, so the accompanying text notes only the main modeling relationships.

4.7.1 Actors and Modeling Scope

The principal human actors of Eduverse are summarized in Table 4.1. Their permissions are role-aware and organization-scoped, which means that the same individual may encounter different functional boundaries depending on the active organization and role.

Table 4.1. Principal Eduverse Use-Case Actors

Actor	Analytical Role in the System
Organization Owner	Governs the organization workspace, manages memberships and roles, configures features and extensions, oversees classes, and monitors organization-level dashboards.
Teacher	Manages instructional workflows such as classes, materials, communication, assignments, exams, grading, live sessions, AI-assisted support, and IDE-enabled learning activities.
Student	Participates in the learning process by joining classes, accessing materials, communicating, submitting assignments, taking exams, reviewing results, joining live sessions, using AI support, and using the IDE where the class enables it.

The mobile companion does not introduce a new human actor, and the AI layer is treated as a supporting service rather than as a primary actor.

4.7.2 Use Case Diagram Set

The following figures constitute the main use-case model of Eduverse. Each diagram corresponds to a functional module, while the project overview figure provides the high-level structure of the platform.

Project Overview

Figure 4.1 presents the principal Eduverse modules and their relationship to the organization owner, teacher, and student. It frames the system as one coordinated educational environment.



Figure 4.1. Eduverse project overview use-case diagram

Authentication and Access

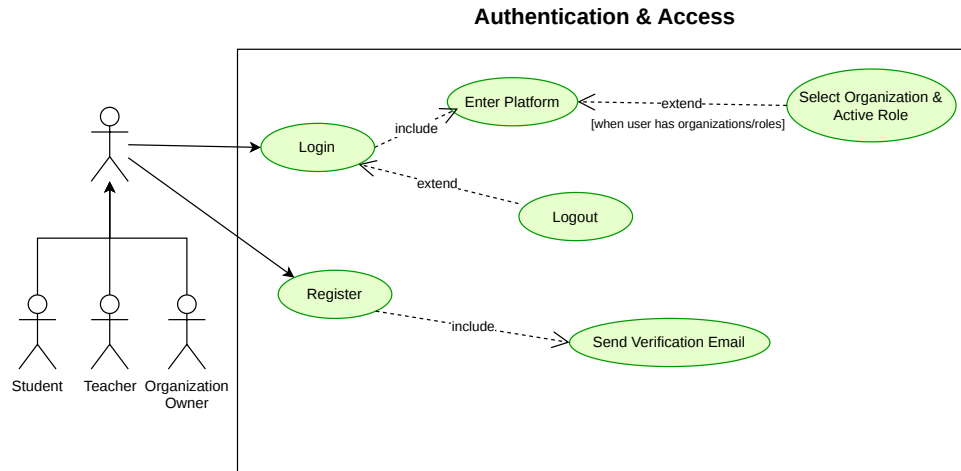


Figure 4.2. Eduverse authentication and access use-case diagram

This diagram shows the public entry workflow and the conditions required before the workspace opens. Register includes Send Verification Email, while Select Organization & Active Role is conditional and applies only when the authenticated user has multiple eligible contexts.

Organization and Role Management

The organization and role management diagram concentrates on institutional governance, including creation, membership control, roles, and configuration. Create Organization includes Apply Organization Feature Preset, which reflects that organizational setup and feature initialization occur together.

Class Management

This diagram positions the class as the main operational unit of the platform. It groups class creation, editing, teacher assignment, membership management, and feature configuration within one controlled instructional context.

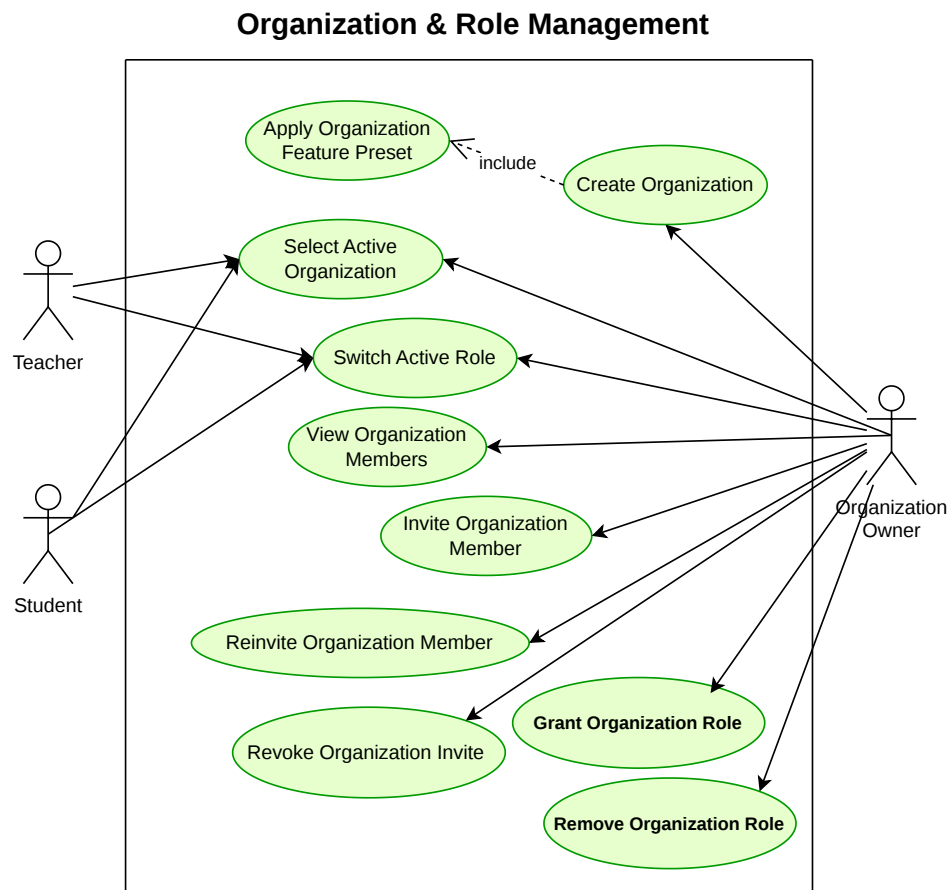


Figure 4.3. Eduverse organization and role management use-case diagram

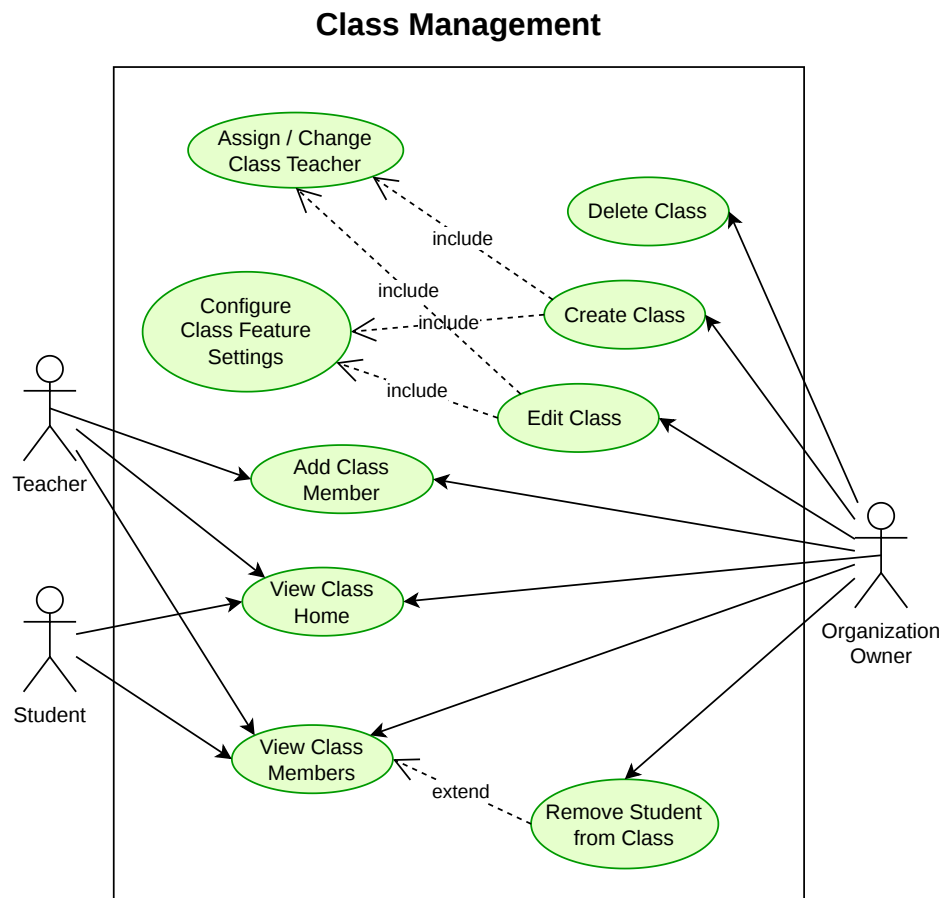


Figure 4.4. Eduverse class management use-case diagram

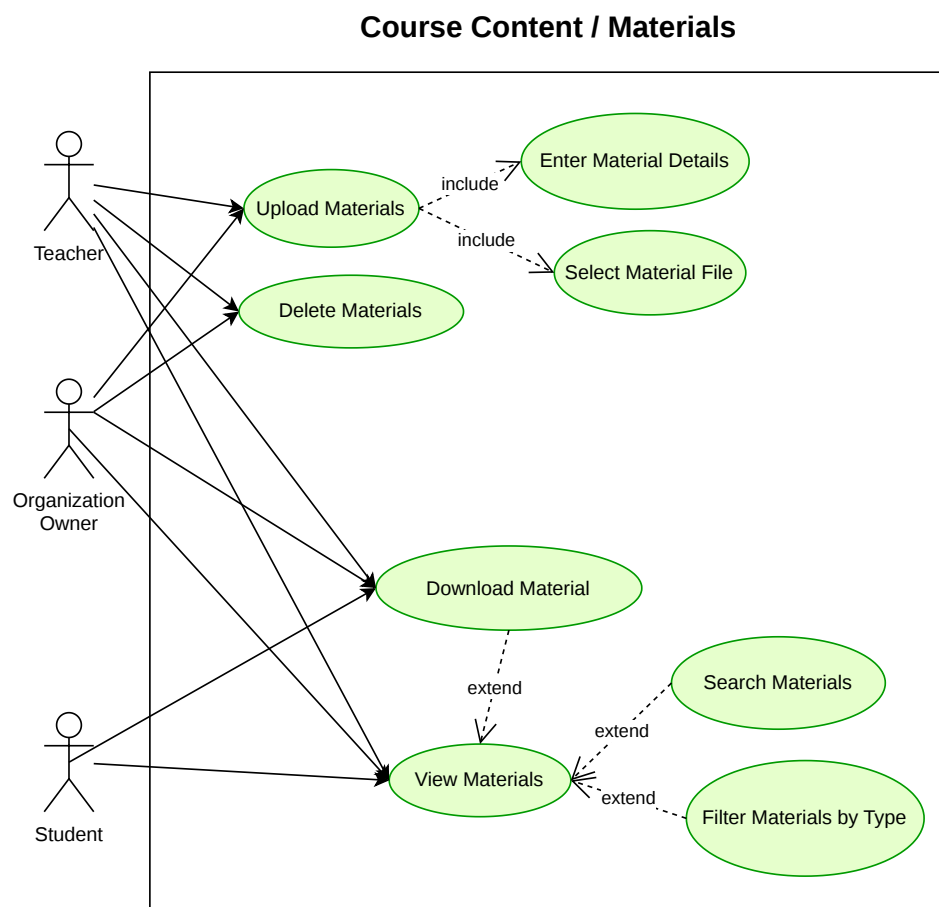


Figure 4.5. Eduverse course content and materials use-case diagram

Course Content and Materials

The materials diagram captures the life cycle of learning resources inside the class context. Upload Materials includes file selection and entry of material details, while search, filter, and download extend the main material-viewing workflow.

Class Communication and Notifications

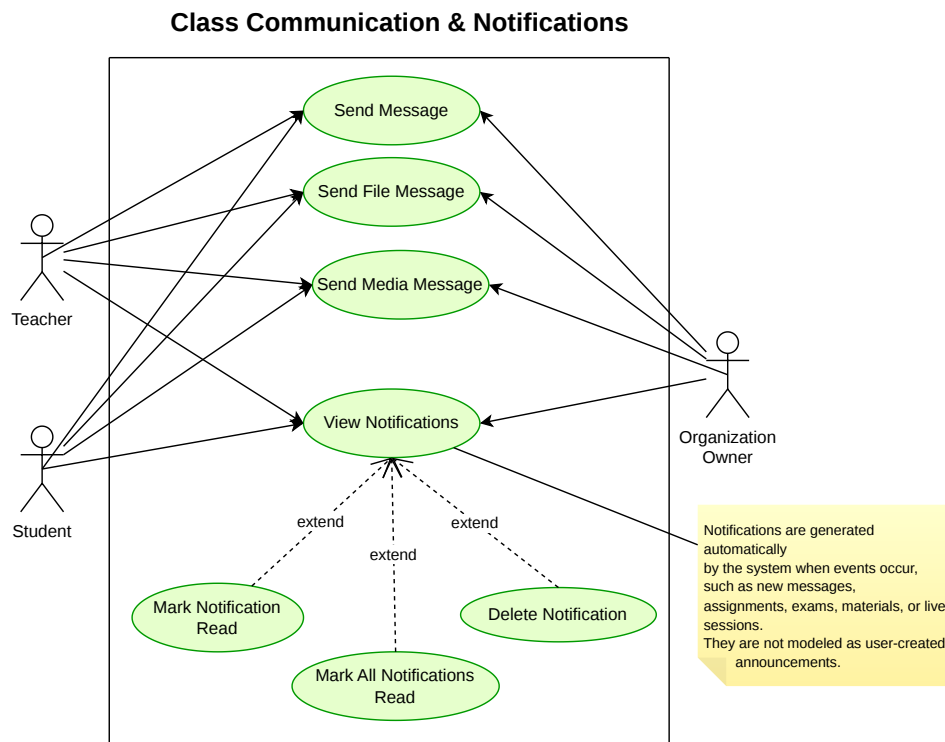


Figure 4.6. Eduverse class communication and notifications use-case diagram

This figure distinguishes direct class interaction from event-driven awareness. Messages are initiated by users, whereas notifications are generated automatically by system events and are followed by read-state or deletion actions within the notification view.

Assignment Management

The assignment diagram represents authoring, publication, submission, grading, and feedback as one connected workflow. Textual submission and file-based submission operate as conditional extensions according to the submission mode allowed for the assignment.

Exam Management and Integrity

This diagram captures the controlled assessment workflow and its integrity checks. Enter Exam Passcode extends Start Exam Attempt only when a passcode is configured, Monitor Exam Attempt includes Review Integrity Events, and Release Exam Results remains an optional step after grading.

Results and Dashboards

The results and dashboards diagram emphasizes differentiated visibility across roles. It combines role-specific dashboards with controlled access to assignment and examination outcomes, including detailed review where policy and permissions permit it.

Live Sessions and Whiteboard

This diagram models the synchronous learning workspace of Eduverse. Media and collaboration features such as microphone, camera, screen sharing, chat, and whiteboard usage extend the main session workflow.

AI Learning Support

The AI learning support diagram positions the AI layer as contextual academic assistance inside the class environment. Summarization, drafting support, and learning help extend study, assignment, examination, and feedback workflows without replacing teacher judgment.

Integrated Coding Lab / IDE Extension

This figure represents the programming-learning environment integrated into Eduverse. File management, writing or editing code, code execution, terminal use, preview, and console or problem viewing all extend the main IDE workspace.

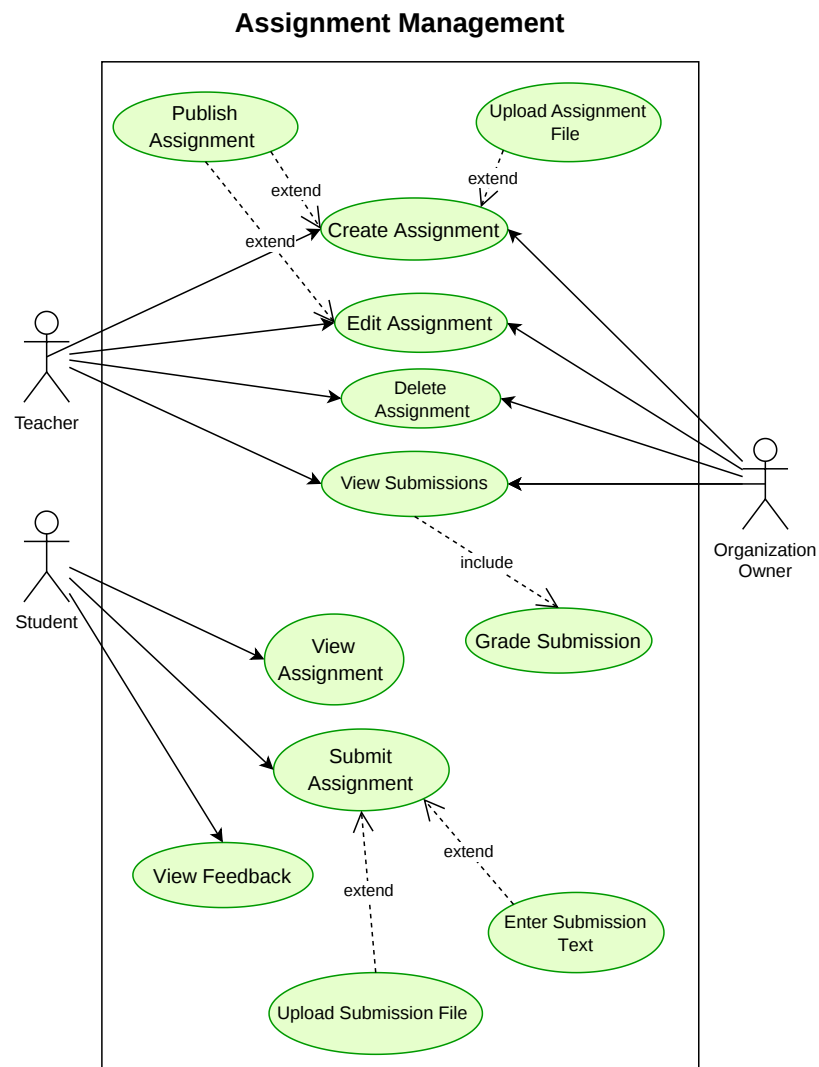


Figure 4.7. Eduverse assignment management use-case diagram

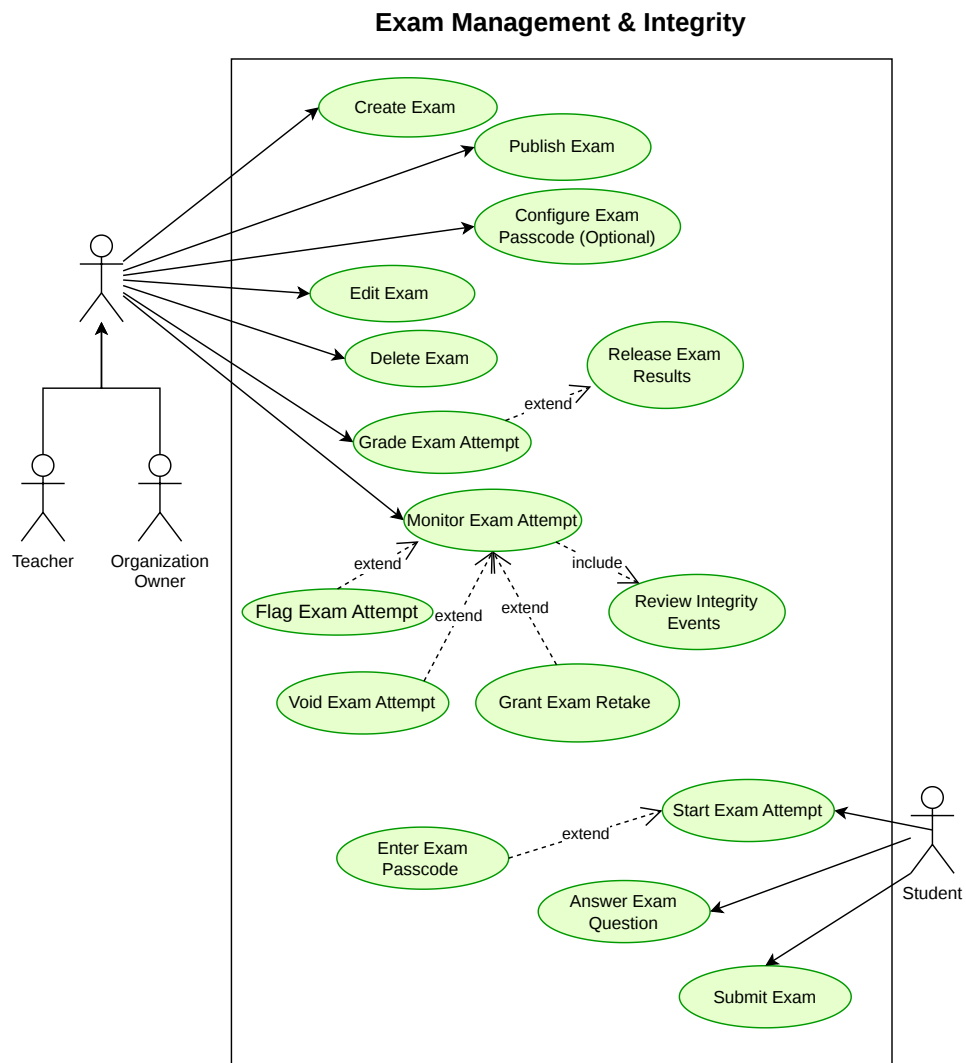


Figure 4.8. Eduverse exam management and integrity use-case diagram

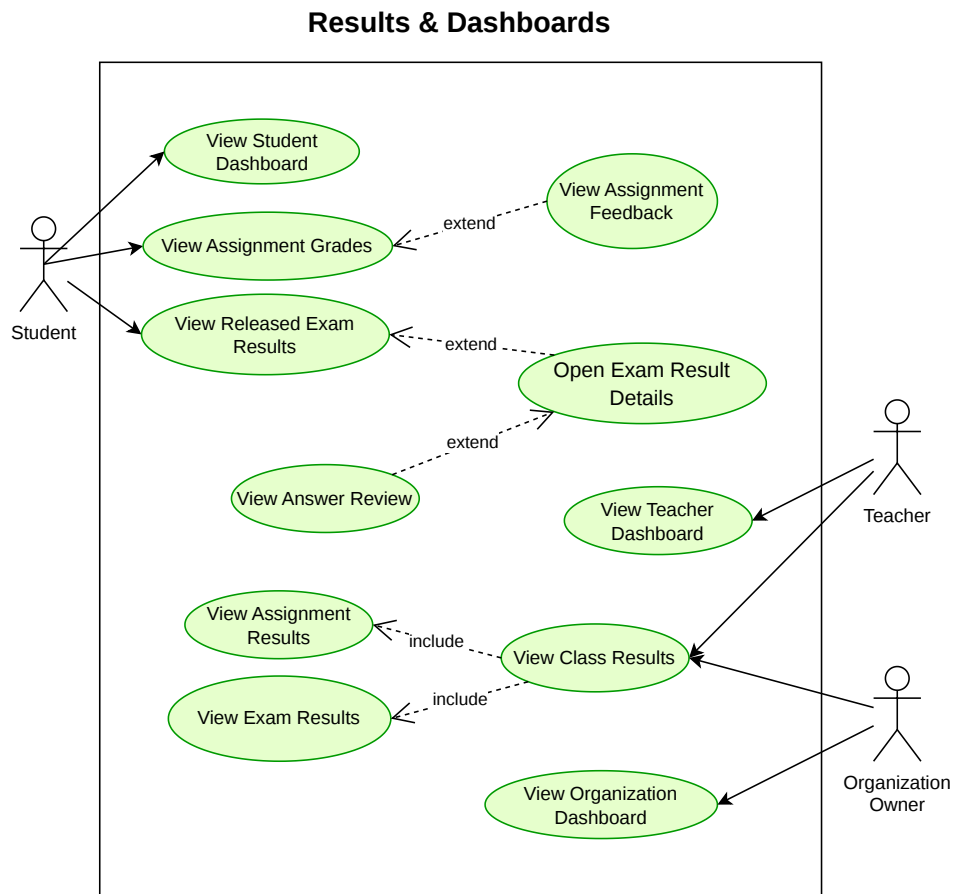


Figure 4.9. Eduverse results and dashboards use-case diagram

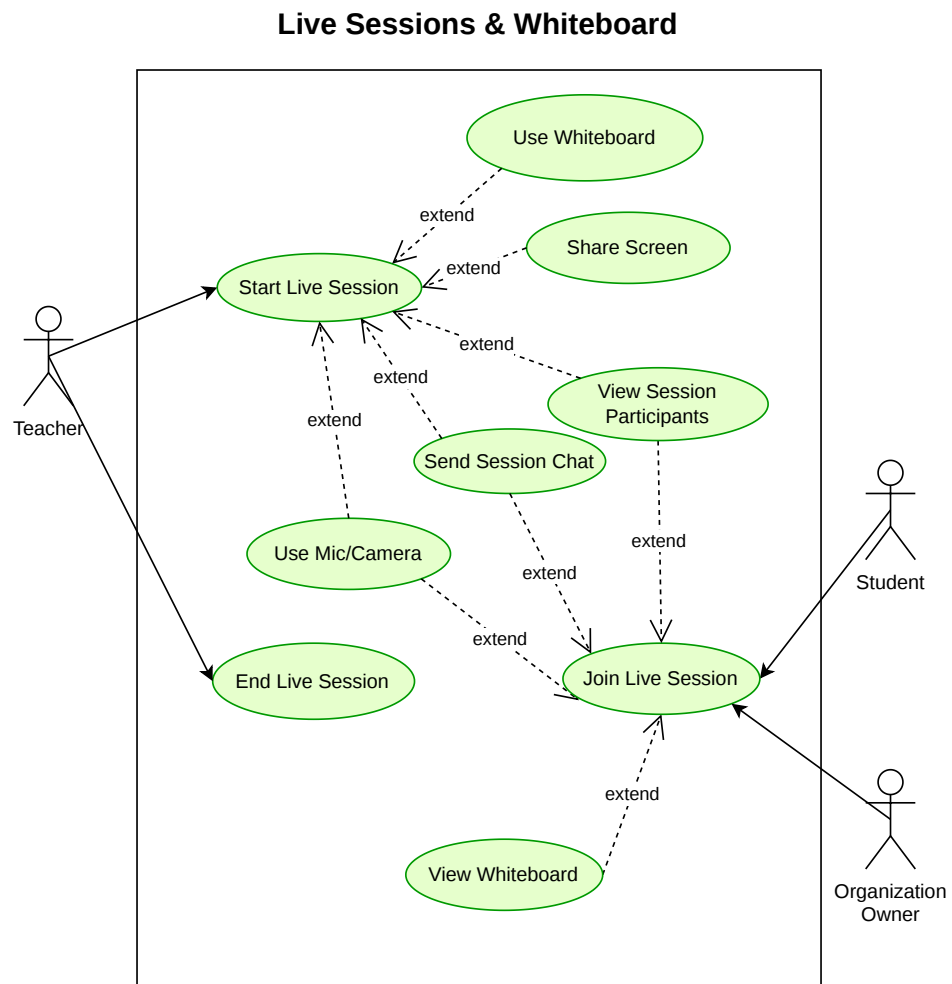


Figure 4.10. Eduverse live sessions and whiteboard use-case diagram

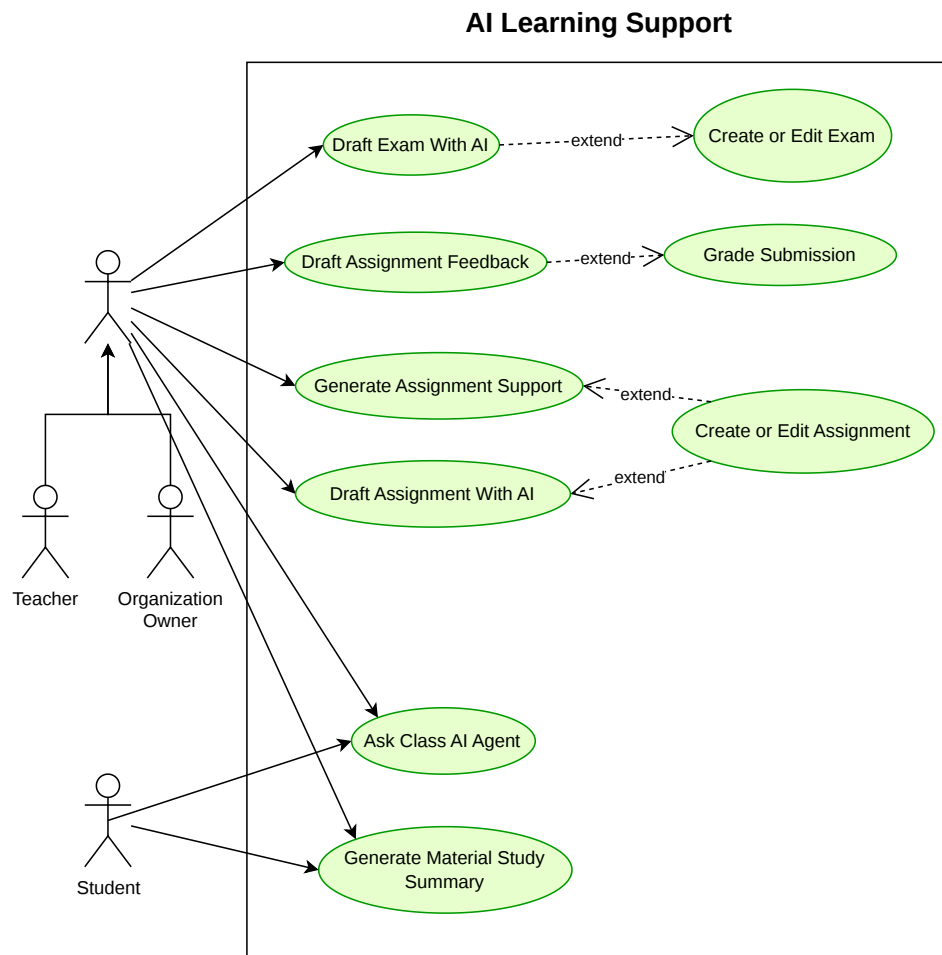


Figure 4.11. Eduverse AI learning support use-case diagram

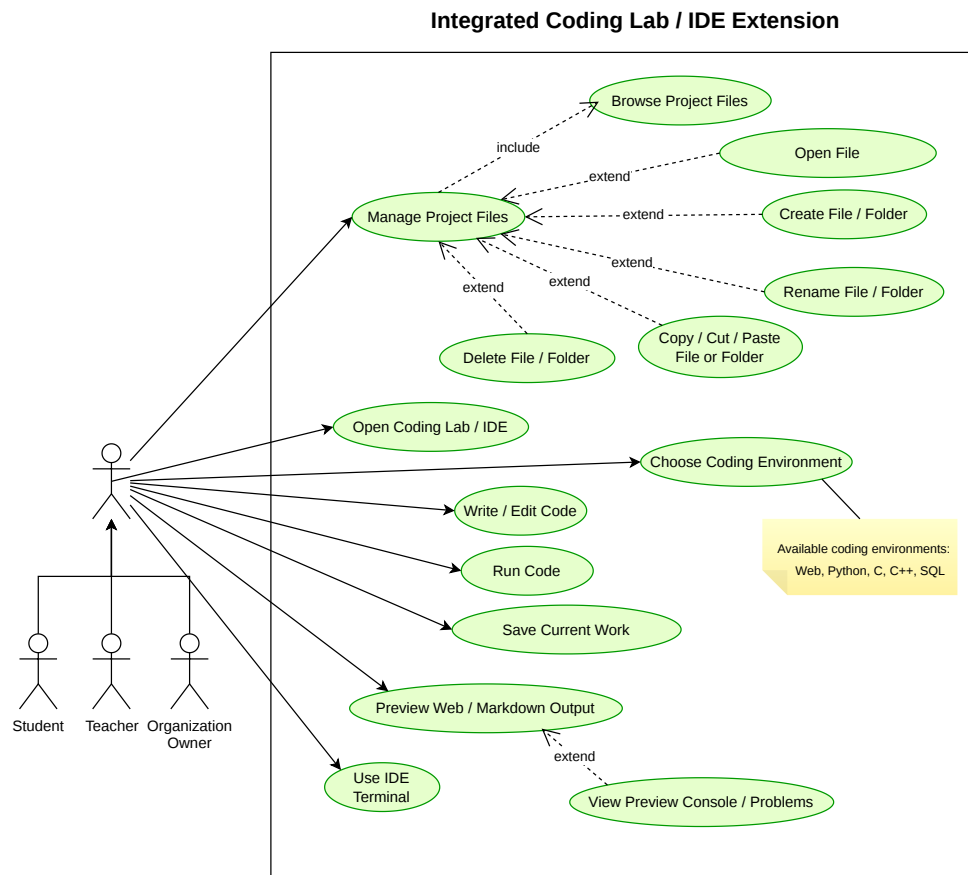


Figure 4.12. Eduverse integrated coding lab / IDE extension use-case diagram

Profile and Help

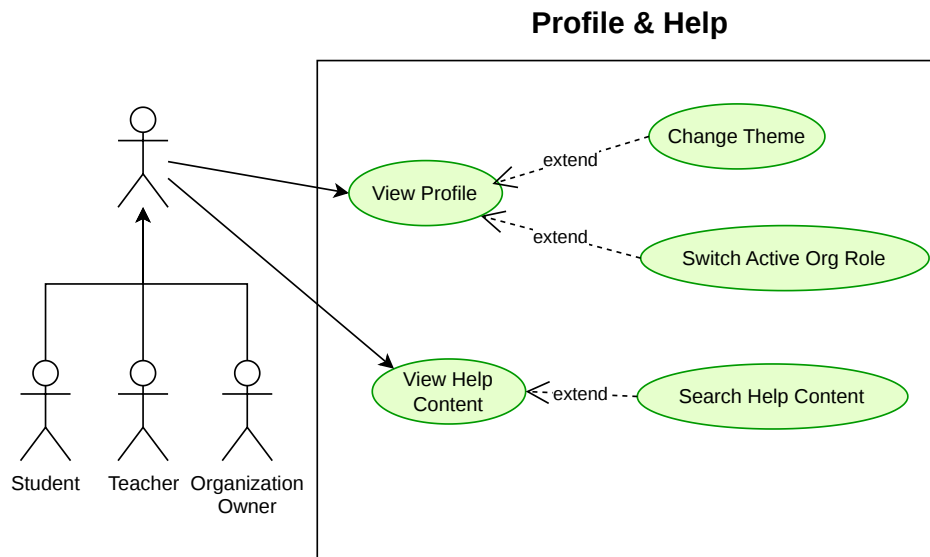


Figure 4.13. Eduverse profile and help use-case diagram

The profile and help diagram keeps the user-support scope intentionally simple. View Profile acts as the base interaction, while theme change, role switching, and help search extend profile or help viewing where appropriate.

4.8 Activity Diagrams

The activity diagrams in this section model the operational workflows that connect user actions, visible system responses, and key decision points inside Eduverse. They intentionally omit low-level implementation detail so that each workflow remains clear at the analytical level.

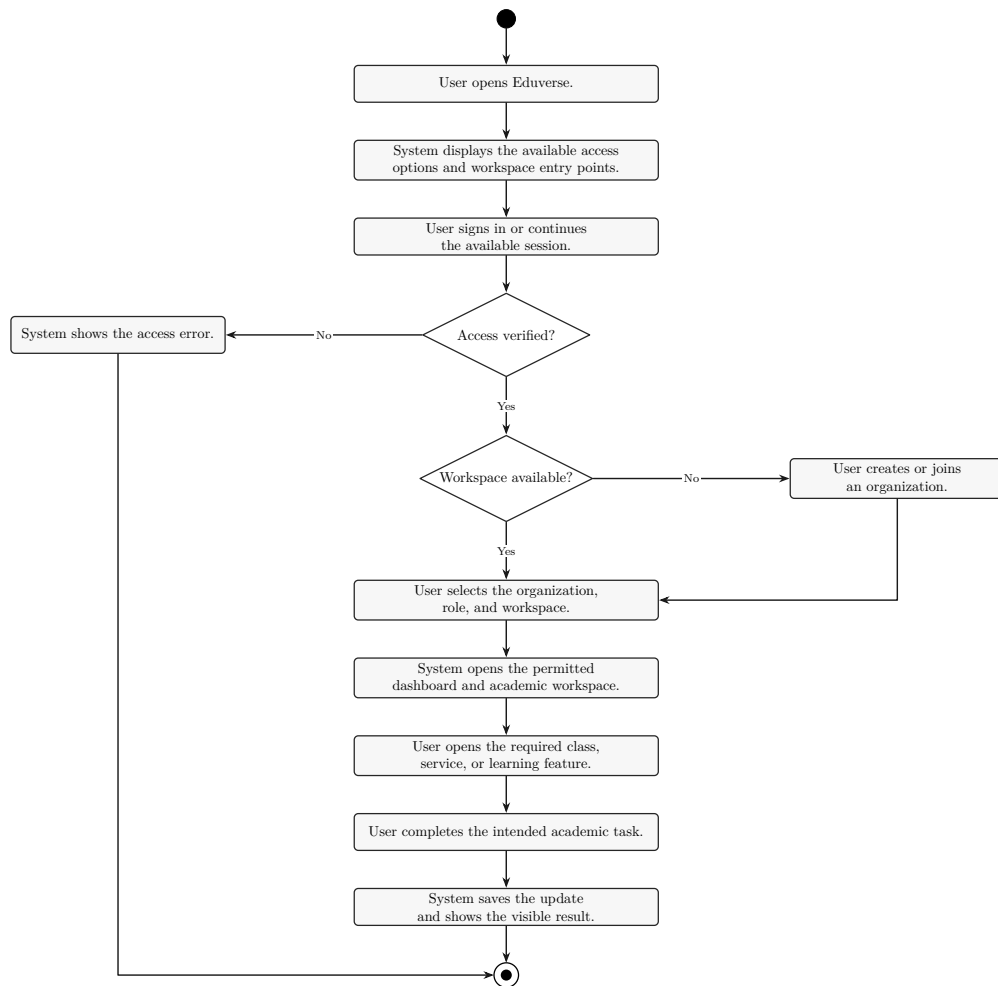


Figure 4.14. Activity diagram representing the overall Eduverse system workflow

This figure presents the top-level flow from platform entry to completion of a visible academic task within the permitted workspace.

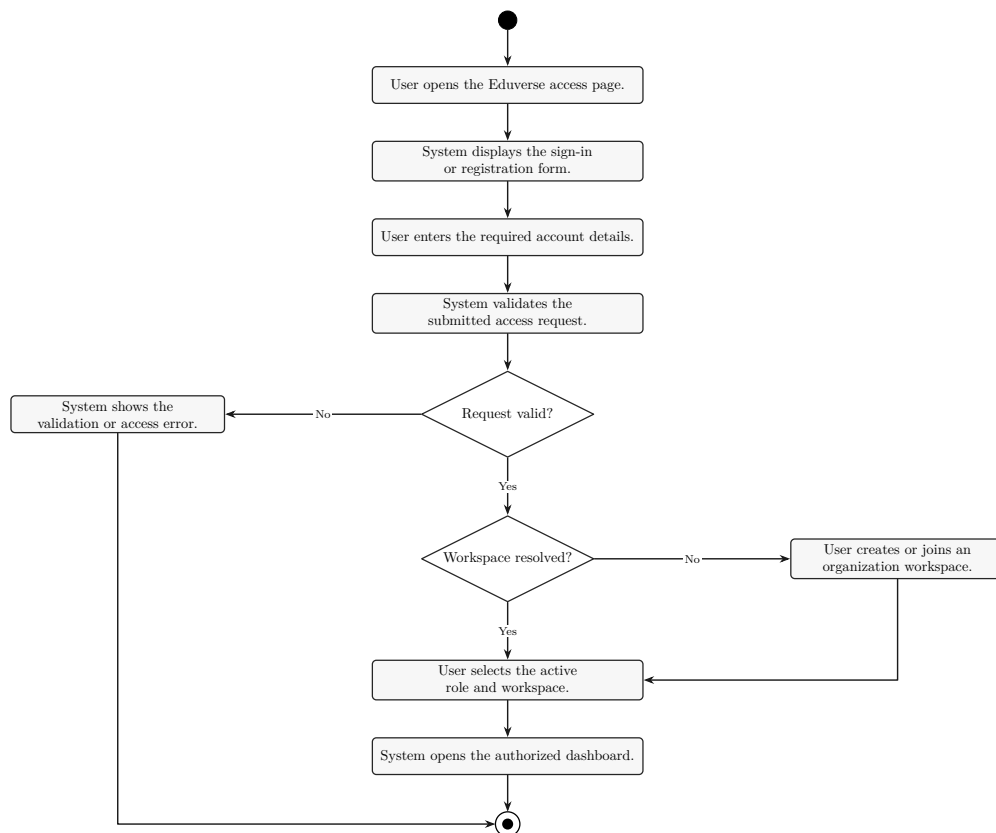


Figure 4.15. Activity diagram representing authentication and workspace access

The authentication workflow links identity validation to the resolution of the active role and workspace.

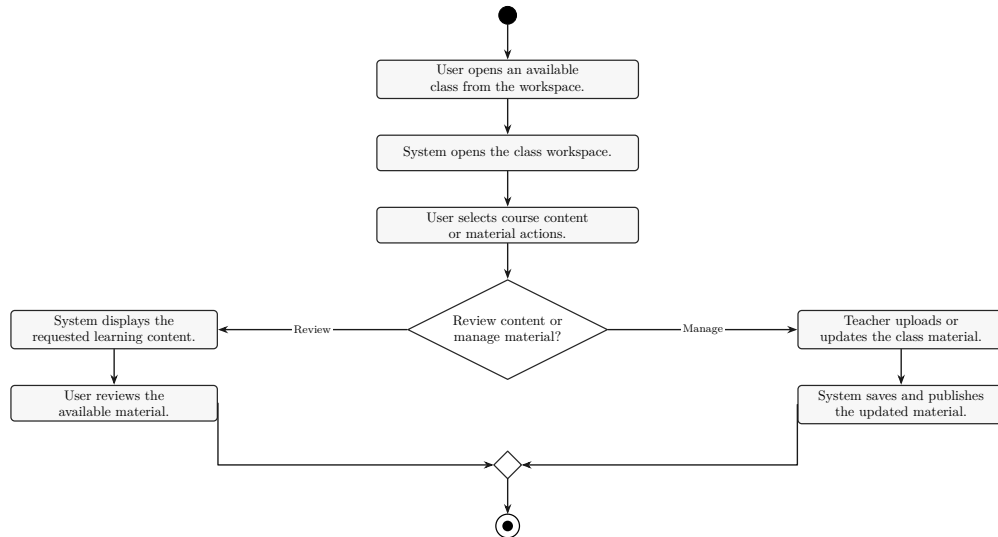


Figure 4.16. Activity diagram representing class access and course content workflow

This workflow shows how class access control leads users into viewing or managing course content according to the active role.

The assignment workflow combines teacher-side publishing and grading with the student-side submission path in one controlled class context.

This diagram captures the examination flow from publication and attempt access to submission, review, and result release.

The live-session workflow emphasizes synchronous participation through session activation, student entry, and controlled session closure.

This workflow focuses on how the AI Agent supports learning interactions through question entry, controlled response generation, and continued use when needed.

This workflow models controlled access to programming and extension tools, including workspace entry, code execution or preview, and optional continued interaction.

Together, these activity models show how Eduverse organizes core academic workflows through connected but controlled interactions. The following chapters build on this analysis to explain the design and implementation decisions in greater technical detail.

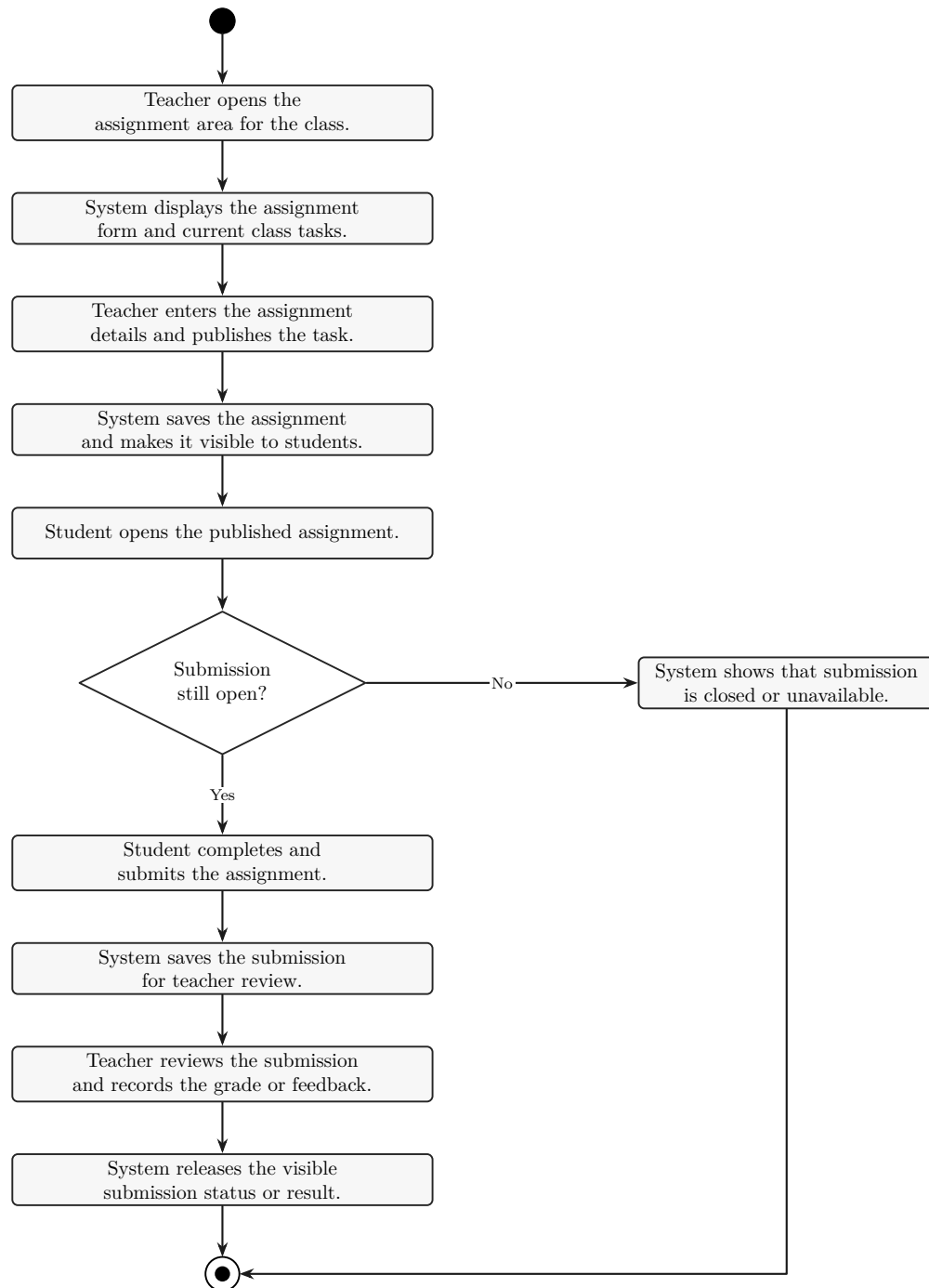


Figure 4.17. Activity diagram representing assignment creation, submission, and grading workflow

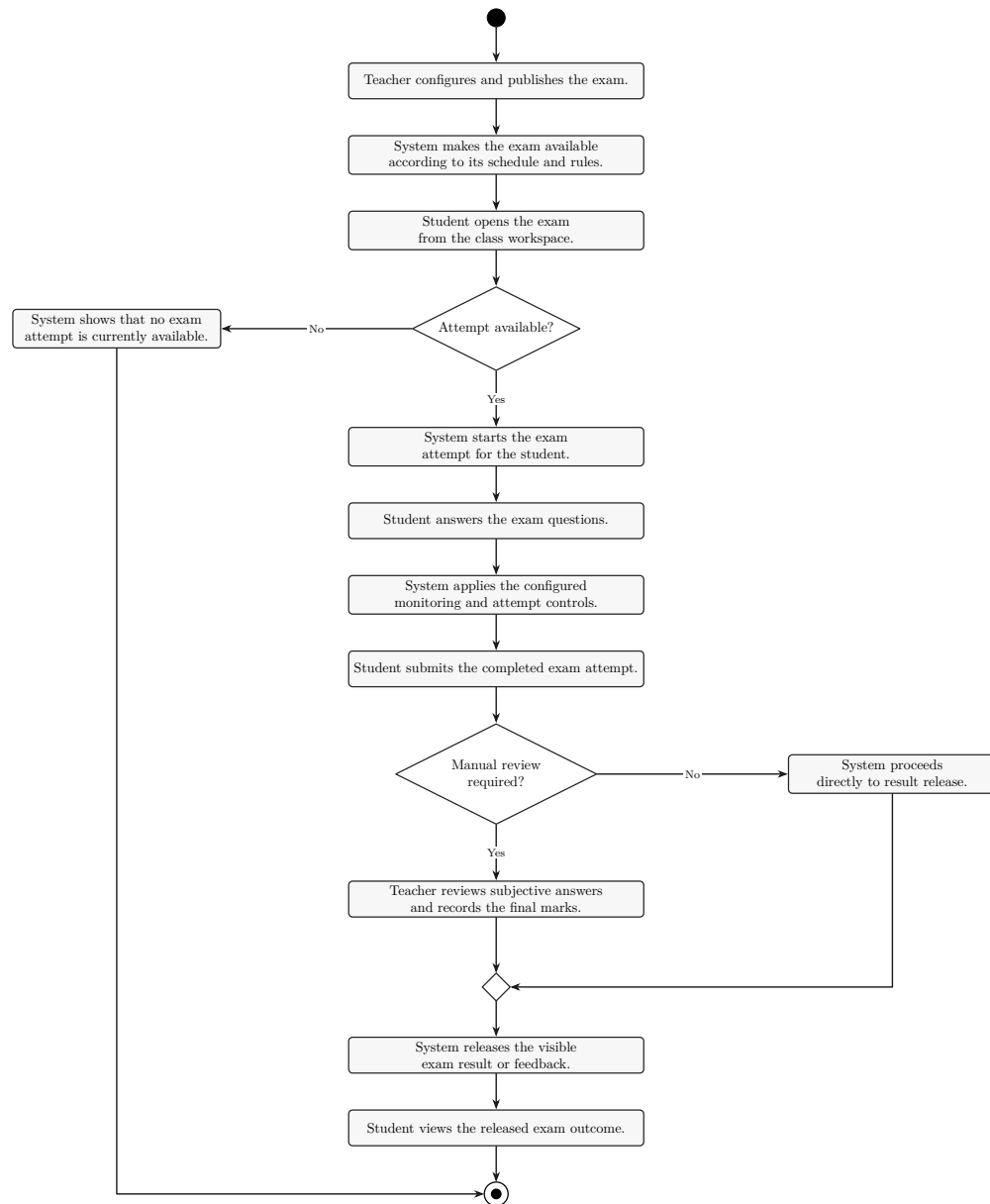


Figure 4.18. Activity diagram representing exam attempt, monitoring, grading, and result release workflow

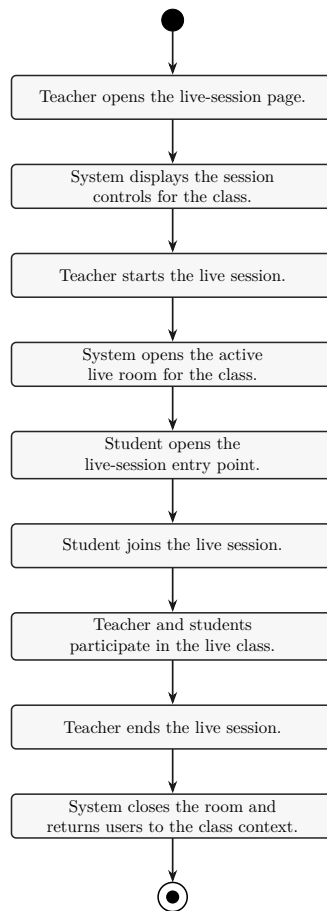


Figure 4.19. Activity diagram representing live session start, join, and end workflow

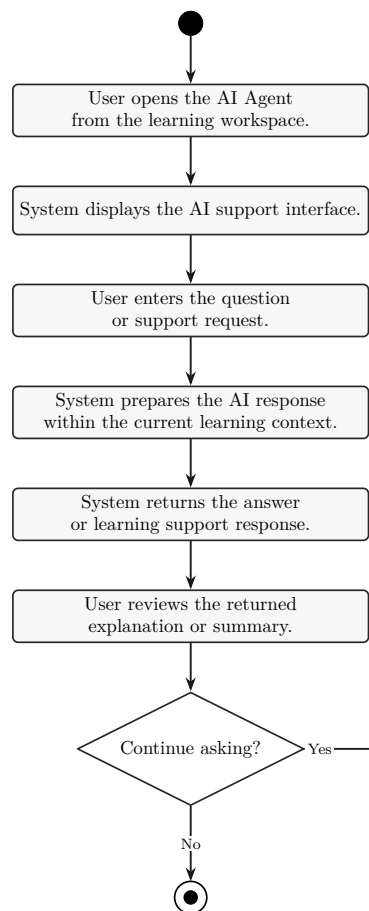


Figure 4.20. Activity diagram representing AI learning support workflow

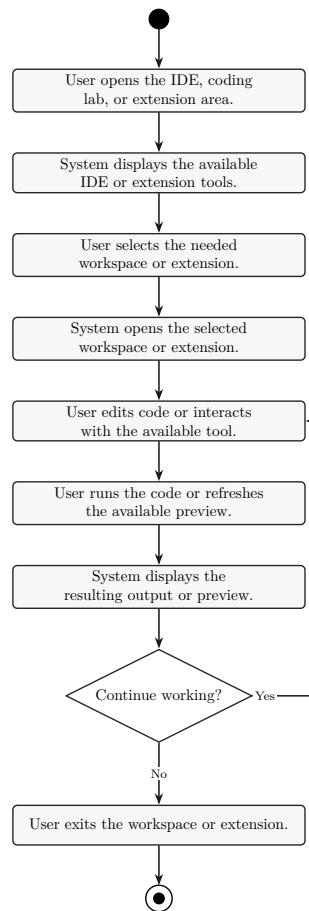


Figure 4.21. Activity diagram representing IDE, coding lab, and extensions workflow

Chapter 5

System Design

TODO: This chapter outline is reserved for a later documentation step.

- 5.1 Introduction
- 5.2 Architectural Design Overview
- 5.3 Data and Storage Design
- 5.4 Interface and Interaction Design
- 5.5 Integration and Security Design

Chapter 6

Software and Engineering Implementation

TODO: This chapter outline is reserved for a later documentation step.

6.1 Introduction

6.2 Web Application Implementation

6.3 Mobile Companion Implementation

6.4 Backend and Service Integration

6.5 Engineering Challenges and Resolutions

Chapter 7

System Testing and Evaluation

TODO: This chapter outline is reserved for a later documentation step.

7.1 Introduction

7.2 Testing Strategy

7.3 Functional Testing

7.4 Non-Functional Evaluation

7.5 Discussion of Results

Chapter 8

Future Work and Planned Features

TODO: This chapter outline is reserved for a later documentation step.

8.1 Introduction

8.2 Mobile Expansion

8.3 AI Agent and IDE Enhancement

8.4 Scalability and Institutional Growth

Chapter 9

Conclusion

TODO: Write the final conclusion after the main report chapters are completed.

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Appendix A

Appendices